vision2learn

Understanding Coding
Level 2

Hints and Tips

Unit 4

Contents

Hints and Tips Unit 4

The information in this guide will be helpful to you when you are ready to complete your assessment:

Unit 4: Understand the processes and practice of coding Level 2

Section 1 – Understand the principles of best practice in coding

Q1. In the box below, describe what is meant by the KISS principle.

This question is quite straightforward and you simply need to describe what the KISS principle is. Think about why it is useful in coding. Try and provide good detail in your response and you may want to give an example of when it could be used and why.

Q2. In the table below, give examples of the disadvantages of writing complicated code.

For this question you need to think of when it would be a disadvantage to write complicated code. You may want to do some research on this to assist you. Try and give at least 2 examples in your answer.

Q3. In the table below:

a. Define what is meant by the single responsibility principle
b. Describe what is meant by separation of concerns
c. Define what is meant by abstraction.

For this question you are being asked to give 2 definitions and also a description so please take care when formulating your responses. For all three you may find it useful to give examples to demonstrate your understanding.

Q4. In the box below, describe what is meant by solid principles.

This is a describe question so try to provide good detail in you answer. Make sure that you cover all 5 of the principles in your response.

Section 2 – Understand the methods of testing code

Q1. In the table below:
a. Explain what is meant by testing code
b. Explain why it is important to test code

This question covers both learning outcomes 1.1 and 1.2. For both elements you are being asked to explain so try to provide good detail in your response for each.

Q2. In the table below, identify methods of testing code.

This is an identify question so you may just list your responses. Try and give as many methods as you can.

Q3. In the table below, identify what is meant by a bug in relation to code.

For this question you need to identify different types of bug and give a summary of the different types. Try to be as detailed as possible in your response to demonstrate your understanding.

Q4. In the box below, describe what is meant by debugging.

This is a describe question so try to provide good detail in your answer. You may want to give example of how debugging is used.

Q5. In the table below, describe the characteristics of a Test Driven Development (TDD).

This is a describe question so you need to provide good detail in your response. Think about what is does and the different steps in the test.

Q6. In the table below, explain the benefits of a Test Driven Development (TDD).

This is an explain question so try and provide good detail in your response. Try and give as many benefits as possible but three at a minimum.

Section 3 - Understand DevOps processes

Q1. In the table below define what is meant by:
a. continuous integration [3.1]
b. continuous delivery [3.2]
c. continuous deployment [3.3]

This question covers learning outcomes 1.1 to 1. 3. For this question you simply need to give a definition of what each of these mean. What is their purpose? You may want to give examples to demonstrate your understanding

Q2. In the table below, identify the steps required for:
• continuous integration
• continuous delivery
• continuous deployment

This is an identify question so you can simply list the steps but make sure that you cover all the steps required for each factor.

Q3. In the box below, identify the differences between:
• continuous integration
• continuous delivery
• continuous deployment. You may find it useful to use an example.

For this question you need to give the difference between all three. When is each one used? As the question suggests you may find it easier to answer the question if you use examples.

Section 4 - Understand the importance of robust coding

Q1. In the box below, explain what is meant by exception handling.

This is an explain question so try and provide good detail in your response. Under what cirmcumstances is exception handling used? You may find it useful to give an example to demonstrate your understanding.

Q2. In the box below, explain what is meant by defensive programming.

This is another explain question so provide good detail in your response. What is defensive programming? What is it designed to do? Try and give an example to demonstrate understanding.